

PRAJJWAL DUTTA

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SUMMARY

Autonomy engineer with dual M.S. from ASU (Robotics and Manufacturing Engineering), specializing in motion planning, MPC/CLF-CBF-QP control, and multi-agent coordination. Experienced building end-to-end ROS2 autonomy stacks integrating planning, state estimation, perception, and control; validated on ground, aerial, and distributed robotic platforms. Proficient in C++, Python, and MATLAB/Simulink; peer-reviewed publication at ICRA 2025.

EDUCATION

Manufacturing Engineering, M.S.

Expected Dec 2026

Arizona State University, Mesa, AZ

GPA: 4.33

Relevant coursework: Convex Optimization, Complex Engineering Analysis

Robotics and Autonomous Systems (Electrical Engineering), M.S.

Aug 2025

Arizona State University, Tempe, AZ

GPA: 3.84

Relevant coursework: Optimal Control, Linear Systems Theory, Robot Kinematics, Space Robotics

TECHNICAL SKILLS

Programming: Python, C++, MATLAB/Simulink, Pandas, NumPy, PLC programming (Ladder Logic, Codesys)

Controls & Autonomy: MPC, Tube-MPC, LQR, CLF-CBF-QP, nonlinear control, trajectory tracking, visual servoing, coverage planning, multi-agent coordination, deep reinforcement learning (SAC)

Planning & Estimation: A*, Hybrid A*, RRT*, motion planning, SLAM, Kalman filtering, particle filtering, sensor fusion, Bayes filtering, state estimation

Robotics Software: ROS2, OpenCV, point cloud processing, RANSAC, ArUco detection, Linux, Docker, Git

Simulation & Validation: SolidWorks, HoloOcean, Gazebo, MuJoCo, CarSim, PX4 SITL, MATLAB/Simulink, HITL testing, sim-to-real deployment

Platforms & Hardware: TurtleBot 4, FANUC LR Mate, UR5, OAK-D RGB-D, Parrot Mambo, Agile X Limo, RosMASTER X3, blimp, custom aerial platform, motion capture (MoCap)

PROFESSIONAL EXPERIENCE

Research Assistant in AMASS Lab | NSF- & NIWC-funded, **Mesa, AZ: Arizona State University** Aug 2024 - Present

- Designed a scalable decentralized multi-robot coordination system using graph-pruning algorithms and CLF-CBF-QP safety-constrained control; validated on 20-agent simulation with 96% faster convergence and 99.3% lower inter-agent communication overhead under degraded comms and external disturbances, achieving 100% network connectivity across all trials.
- Built a warehouse autonomous navigation stack benchmarking motion planning algorithms (A*, Hybrid A*, RRT*) against trajectory-tracking controllers (MPC, LQR, Pure Pursuit, Stanley, PID); deployed and validated on an Agile X Limo ground robot with MoCap-based ground-truth verification, achieving best-case 1.28 m max lateral deviation at operating speed.
- Developed a distributed connectivity-maintenance control law for multi-vehicle fleets; designed and built a custom aerial platform for hardware validation and conducted experimental flight tests confirming sim-to-real transfer of the control framework.

RESEARCH & PUBLICATIONS

Distributed Control and Communication Framework for Connectivity Preservation in Flow Driven Robot Fleets
IROS 2026

Prajjwal Dutta, Geoffrey Hollinger, Xi Yu. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. (*distributed connectivity-preserving control for robot fleets in flow fields*)

Connectivity-Aware Coordination of AUV Fleets in Dynamic Underwater Environments

ICRA 2025

Prajjwal Dutta, Jnaneshwar Das, Xi Yu. RoboAcoustics Workshop @ IEEE ICRA 2025. (*distributed multi-agent control under bandwidth-constrained communication*)

Connectivity-Aware Coordination and Control of Autonomous Underwater Vehicles in Environment Dynamics

Prajjwal Dutta. M.S. Thesis, Arizona State University, 2025. ProQuest ID: 32122670.

PROJECTS

Tube-MPC Lateral Path Tracking with CarSim-MATLAB/Simulink Co-Simulation

Spring 2026

- Implemented a Tube-MPC lateral controller combining nominal MPC trajectory tracking with discrete-time LQR disturbance rejection; Simulink validation reduced max lateral error from 18.80 m to 0.032 m.

- Validated robustness in CarSim/Simulink high-fidelity vehicle co-simulation at 54 km/h: 0.29 m max lateral deviation over a 30 s run under realistic vehicle dynamics.

TurtleBot 4 Autonomous Human-Following System

Spring 2026

- Built a TurtleBot 4 human-following system integrating OAK-D ArUco 6-DoF pose estimation (OpenCV solvePnP), dual Kalman/particle filter (300-particle) state estimators, PD velocity controller with 3-state FSM, and LiDAR collision avoidance.
- Validated on hardware: Kalman filter achieved 0.04 m mean tracking error with zero divergence vs. 0.12 m with divergence at 12 s for the particle filter.

ROS2 Autonomy Stack: Perception, Localization & Energy-Aware Planning

Spring 2026

- Developed a full ROS2 autonomy stack with three integrated modules: perception (OAK-D RANSAC cylinder detection, 50-200 ms/frame) and localization (3D Bayes filter, $80 \times 80 \times 72$ pose grid, <0.6 m convergence within 3 observations).
- Planning module: A* on a downsampled 80×80 grid with Bresenham line-of-sight pruning reduced waypoints from 108 to 11 ($67 \times$ speedup, 89.8% reduction); Turn-Go-Turn state machine (IDLE, ROTATE, DRIVE) minimized unnecessary rotations for $\sim 90\%$ total energy reduction and 100% collision-free navigation; validated with 34 unit tests.

ROSMaster X3 Digital Twin & Autonomous Warehouse Navigation

Spring 2026

- Developed a ROS digital twin of the ROSMASTER X3 mecanum-drive robot in a Gazebo warehouse environment; implemented a custom A* global planner (C++) with AMCL particle-filter localization, DWA local planner, and layered costmaps for autonomous point-to-point navigation.
- Performed SLAM (gmapping and Cartographer) to generate and save warehouse maps; integrated YOLOv5 traffic sign detection for sign-conditioned navigation behavior.
- Validated the full autonomous navigation stack on a physical ROSMASTER X3 (Jetson) via SSH.

PLC Tank Level Control: Codesys and Factory IO Co-Simulation

2026

- Programmed a water level tank controller in Codesys (IEC 61131-3 Function Block Diagram); designed custom function blocks with configurable TON/TOF timers for fill/drain valve sequencing and hysteresis-based set-point level control logic.
- Connected the Codesys runtime to Factory IO industrial co-simulation via virtual I/O mapping; validated closed-loop tank level control with sensor-triggered valve actuation and timer-sequenced fill/drain transitions in a real-time virtual process environment.

ROS2 Visual Servoing Pipeline: FANUC LR Mate

Fall 2025

- Developed a ROS2 visual servoing pipeline for the FANUC LR Mate; camera-based target tracking maps 2D image detections to 3D workspace goals via geometric inverse kinematics with elbow-up collision avoidance.
- Enabled autonomous end-effector positioning and target following without explicit goal programming.

Autonomy Modules: Optimal Control, Coverage Planning & UAV Precision Landing

Fall 2025

- Implemented LQR cart-pole stabilization under seismic disturbances (72% peak-force reduction, 3.1 to 1.1 m displacement) and a boustrophedon coverage navigator (<0.2 unit avg cross-track error).
- Built a PX4 SITL UAV using RTAB-Map SLAM and RGB-D sensing for autonomous target detection, dimension estimation, and precision landing.

Scalable Multi-Robot Flocking and Coordination (50-Agent Simulation)

Fall 2025

- Implemented a 50-robot distributed coordination simulation in Simulink with LiDAR-based local sensing, flocking control algorithms, and reactive obstacle avoidance.
- Demonstrated scalable convergence to coordinated formation from arbitrary initial configurations.

Autonomous Blimp Navigation Stack with Sensor Fusion and Real-Time Detection

Spring 2025

- Designed custom blimp airframe (SolidWorks), fabricated the platform, and designed a custom integration PCB in **Altium Designer** with dedicated ports for ESC, IMU, barometer, and JST power connectors; implemented **I2C** for IMU sensor data and **UART** for ESC motor control on a Raspberry Pi and ESP32 compute stack; debugged hardware bring-up with oscilloscope and logic analyzer.
- Built a ROS2 navigation stack integrating YOLOv5 detection (96% mAP@0.5, under 100 ms CPU inference) with Kalman-filter sensor fusion (200 Hz IMU and 30 fps camera) for real-time target tracking; platform served as a hardware testbed for validating multi-agent control algorithms in sim-to-real experiments.

Bio-Inspired Flapping-Wing Robot Design and Simulation

Fall 2024

- Designed and fabricated a flapping-wing robot with gear-driven four-bar linkage and adjustable-span wings; created **SolidWorks** CAD assemblies for all mechanical components prior to fabrication.

- Validated dynamic behavior using MuJoCo rigid-body simulation against the physical prototype via Jacobian kinematics analysis.

Won 1st Place: MATLAB Mini Drone Competition

Spring 2024

- Led a 4-person team to 1st place; engineered a Simulink sim-to-real autonomy pipeline with SLAM and path following.
- Deployed and validated the full autonomy solution on physical hardware.

Perception-Driven Manipulation Robot: Tic-Tac-Toe with MyCobot 280M5

Fall 2023

- Built a full manipulation pipeline on the MyCobot 280M5: ArUco-calibrated eye-to-hand camera transforms pixel coordinates to 3D workspace positions; color segmentation detects board state (blue vs. green blocks); alpha-beta pruning AI selects moves; IK-computed joint trajectories execute suction-pump picks via serial protocol.
- Validated competitive gameplay on physical hardware; MATLAB/Simulink digital twin synchronized joint trajectories on a URDF model and the physical arm simultaneously for a letter-writing demonstration.

UR5 Manipulation Programming

Fall 2023

- Programmed UR5 manipulator (PolyScope, RoboDK, URScript) for multi-task manipulation: precision waypoint execution and pick-and-place.
- Completed assembly tasks with torque-controlled fastening (3 Nm).

SAC Deep Reinforcement Learning for Continuous Robot Control

Spring 2025

- Implemented Soft Actor-Critic (SAC) with dynamic entropy regularization in MATLAB Reinforcement Learning Toolbox; designed a 22-dimensional observation vector (joint kinematics, quaternion plate orientation, ball states) and dual Gaussian actor-critic neural networks for 2-DOF continuous torque control; built Simscape Multibody physics environment for sim-validated training.
- Trained 6,000 episodes on RTX 4090 achieving 660 episode reward; trained stochastic policy stabilizes ball from diverse initial conditions via multiple viable trajectories.